

Note to Prof. Robert Xiao

2026-01-29

Draft message

Subject: 2048 AI port update (TypeScript/Angular) — results & questions

Hi Prof. Xiao,

I'm a developer working on a TypeScript/Angular port of your 2048 AI (based on the classic C++ expectimax implementation you published). I just wanted to share that the port is working well and running surprisingly fast in JS, largely due to the bitboard + row-lookup structure you used.

We mirrored the core heuristics and weights (empty cells, merges, monotonicity, sum penalty, and corner max bonus), and the expectimax tree with 0.9/0.1 spawn probabilities. We're now seeing consistent 8192/16384 runs, with some 32768 outcomes, especially at higher depth caps.

If you're interested, I'd love to share more details or ask a couple of questions about your heuristic tuning (especially around the CMA-ES weight optimization) and any late-game stability tricks you found effective.

Thanks for publishing such a clear and elegant AI — it made a faithful port possible.

Best regards,

[Your Name]

Screenshots

32768 tile achieved (WASM)

Runs history showing 32768 result



Figure 1: 32768 run

Score 585680 Best TS 20632 Best WASM 585680 Home Runs				
AI Run History				
Crossed 387000: 2/7 (28.6%)		Export TS Runs CSV	Export WASM Runs CSV	Export Config CSV
Clear History				
Date/Time	Score	Max Tile	Depth	Top Tiles
1/29/2026, 11:50 AM	78376	4096	3	4096, 2048, 1024
1/29/2026, 1:00 PM	340280	16384	4	16384, 8192, 2048, 1024
1/29/2026, 11:59 AM	290440	16384	5	16384, 4096, 2048, 1024
1/29/2026, 2:50 PM	585680	32768	2	32768, 8192, 2048, 1024

Figure 2: Runs history